

# ★ ★ SOLUTIONS ★ ★



## Tom Price Senior High School

*An Independent Public School*

### Task Cover Sheet

#### Yr 8 Mathematics: INVESTIGATION - VALIDATION

|                                                |                                         |
|------------------------------------------------|-----------------------------------------|
| <b>Teachers Name:</b>                          | <b>Students Name:</b>                   |
| <b>Date:</b>                                   | <b>TOTAL MARKS:</b> / 48                |
| <b>Subject Foci (Outcomes) and background:</b> |                                         |
| Investigation assessing basic skills in:       |                                         |
| - Fractions                                    | - Percentages                           |
| - Decimals                                     | - Integers                              |
| - Area                                         |                                         |
| <b>Conditions:</b>                             |                                         |
| <b>Assistance</b>                              | <b>Calculators are permitted, Ruler</b> |
| <b>Time Allocated</b>                          | <b>60 mins</b>                          |

## Percentages

We are designing a newsletter for our maths department. We don't have a budget to produce the newsletter so we will need to get advertising to cover the cost of producing the newsletter.

You have been given a copy of the template that we are planning on using. It is your job to calculate the percentage of advertising that is in the proposed newsletter.

1. Estimate the percentage of advertising on each page and use this to find the mean. Complete the table.

| Page No        | Estimate % Advertising |
|----------------|------------------------|
| 1              |                        |
| 2              |                        |
| 3              |                        |
| 4              |                        |
| MEAN (AVERAGE) |                        |

/ 4 marks

2. Calculate the area ( $\text{cm}^2$ ) of each of the advertisements on each page. Show the calculation used and the area of each advertisement in the spread.

**Record the areas in the table and calculate the total area of advertisement for each page.**

| Page number 1                 | Advertisement Area ( $\text{cm}^2$ )   |
|-------------------------------|----------------------------------------|
| Advertisement 1               | $11 \times 4.5 = 49.5 \text{cm}^2$     |
| Advertisement 2               | $10.5 \times 4.5 = 47.25 \text{cm}^2$  |
| Advertisement 3               | $4.5 \times 12 = 54 \text{cm}^2$       |
| <b>Total Advertising Area</b> | <b><math>150.75 \text{cm}^2</math></b> |

/ 7 marks

| Page number 2                 | Advertisement Area ( $\text{cm}^2$ )  |
|-------------------------------|---------------------------------------|
| Advertisement 1               | $4.5 \times 7.5 = 33.75 \text{cm}^2$  |
| Advertisement 2               | $15.5 \times 2.2 = 34.1 \text{cm}^2$  |
| Advertisement 3               | $4.5 \times 11.5 = 51.75 \text{cm}^2$ |
| <b>Total Advertising Area</b> | <b><math>119.6 \text{cm}^2</math></b> |

/ 7 marks

| Page number 3                 | Advertisement Area ( $\text{cm}^2$ )  |
|-------------------------------|---------------------------------------|
| Advertisement 1               | $10.5 \times 9 = 94.5 \text{cm}^2$    |
| Advertisement 2               | $4.5 \times 13.5 = 60.75 \text{cm}^2$ |
| Advertisement 3               | $4.5 \times 7.5 = 33.75 \text{cm}^2$  |
| <b>Total Advertising Area</b> | <b><math>189 \text{cm}^2</math></b>   |

/ 7 marks

| Page number 4                 | Advertisement Area ( $\text{cm}^2$ )  |
|-------------------------------|---------------------------------------|
| Advertisement 1               | $3.5 \times 10.3 = 36.05 \text{cm}^2$ |
| Advertisement 2               | $4.5 \times 9 = 40.5 \text{cm}^2$     |
| Advertisement 3               | $8 \times 10.3 = 82.4 \text{cm}^2$    |
| <b>Total Advertising Area</b> | <b><math>158.95</math></b>            |

/ 7 marks

3. Determine the Area of the whole page showing your calculation.

$$15.5\text{cm} \times 21.5\text{cm} = 333.25\text{cm}^2$$

/ 2 marks

4. Determine the percentage of advertising on each page showing your calculations.  
(Round to 2 decimal places)  
Then determine the mean percentage advertising for the whole newsletter.

| Page No. | Calculation                              | Percentage advertising per page |
|----------|------------------------------------------|---------------------------------|
| 1        | $\frac{150.75}{333.25} \times 100$       | 45.24%                          |
| 2        | $\frac{119.6}{333.25} \times 100$        | 35.89%                          |
| 3        | $\frac{189}{333.25} \times 100$          | 56.71%                          |
| 4        | $\frac{158.95}{333.25} \times 100$       | 47.70%                          |
| Mean     | $(45.24 + 35.89 + 56.71 + 47.70) \div 4$ | 46.39%                          |

/ 10 marks

5. Compare your estimate to the actual percentage advertising space. How could you have improved your estimate?

54%

/ 2 marks

6. What fraction of the newsletter remains for us to send out all of the maths news each term?

about 54%  
exact 53.61%

/ 2 marks

## Advertising 1

49.5cm<sup>2</sup>

4.5cm

11cm

# Newsletter Title

## LEAD STORY HEADLINE

This story can fit 175-225 words.

The purpose of a newsletter is to provide specialized information to a targeted audience. Newsletters can be a great way to market your product or service, and also create credibility and build your organization's identity among peers, members, employees, or vendors.

First, determine the audience of the newsletter. This could be anyone who might benefit from the information it contains, for example, employees or people interested in purchasing a product or requesting your service.

You can compile a mailing list from business reply cards, customer information sheets, business cards collected at trade shows, or membership lists. You might consider purchasing a mailing list from a company.

If you explore the Publisher catalog, you will find many publications that match the style of your newsletter.

Next, establish how much time and money you can spend on your newsletter. These factors will help determine how frequently you publish the newsletter and its length. It's recommended that you publish your newsletter at least quarterly so that it's considered a consistent source of information. Your customers or employees will look forward to its arrival.

## SECONDARY STORY HEADLINE

This story can fit 75-125 words.

Your headline is an important part of the newsletter and should be considered carefully.

In a few words, it should accurately represent the contents of the story and draw readers into the story. Develop the headline before you write the story. This way, the headline will help you keep the story focused.

Examples of possible headlines include Product Wins Industry Award, New Product Can Save You Time!, Membership Drive Exceeds Goals, and New Office Opens Near You.

## Advertising 2

10.5cm

4.5cm

47.25cm<sup>2</sup>

## Advertising 3

54cm<sup>2</sup>

12cm

4.5cm

### INSIDE THIS ISSUE

|                    |   |
|--------------------|---|
| Inside Story ..... | 2 |
| Inside Story ..... | 2 |
| Inside Story ..... | 3 |
| Inside Story ..... | 3 |
| Inside Story ..... | 4 |
| Inside Story ..... | 4 |
| Inside Story ..... | 4 |

### SPECIAL POINTS OF INTEREST

- Briefly highlight your point of interest here.
- Briefly highlight your point of interest here.
- Briefly highlight your point of interest here.
- Briefly highlight your point of interest here.

## Advertising 4

33.75cm<sup>2</sup>

1.5

4.5

### INSIDE STORY HEADLINE

This story can fit 150-200 words.

One benefit of using your newsletter as a promotional tool is that you can reuse content from other marketing materials, such as press releases, market studies, and reports.

While your main goal of distributing a newsletter might be to sell your product or service, the key to a successful newsletter is making it useful to your readers.

A great way to add useful content to your newsletter is to develop and write your own articles, or include a calendar of

upcoming events or a special offer that promotes a new product.

You can also research articles or find "filler" articles by accessing the World Wide Web. You can write about a variety of topics but try to keep your articles short.

Much of the content you put in your newsletter can also be used for your Web site. Microsoft Publisher offers a simple way to convert your newsletter to a Web publication. So, when you're finished writing your newsletter, convert it to a Web site and post it.

## Advertising 5

34.1cm<sup>2</sup>

15.5

3.0cm<sup>2</sup>

2.2

## Advertising 6

4.5

51.75cm<sup>2</sup>

11.5

### INSIDE STORY HEADLINE

This story can fit 100-150 words.

The subject matter that appears in newsletters is virtually endless. You can include stories that focus on current technologies or innovations in your field.

You may also want to note business or economic trends, or make predictions for your customers or clients.

If the newsletter is distributed internally, you might comment upon new procedures or improvements to the business. Sales figures or earnings will show how your business is growing.

Some newsletters include a column that is updated every issue, for instance, an advice column, a book review, a letter from the president, or an editorial. You can also profile new employees or top customers or vendors.

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## Advertising 7

94.5cm<sup>2</sup>

a

10.5

## Advertising 8

60.75cm<sup>2</sup>

13.5

4.5

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### INSIDE STORY

## Advertising 9

33.75cm<sup>2</sup>

7.5

4.5

### YOUR ORGANIZATION

This would be a good place to insert a short paragraph about your organization. It might include the purpose of the organization, its mission, founding date, and a brief history. You could also include a brief list of the types of products, services, or programs your organization offers, the geographic area covered (for example, western U.S. or European markets), and a profile of the types of customers or members served.

It would also be useful to include a contact name for readers who want more information about the organization.

### BACK PAGE STORY HEADLINE

This story can fit 175-225 words.

If your newsletter is folded and mailed, this story will appear on the back. So, it's a good idea to make it easy to read at a glance.

A question and answer session is a good way to quickly capture the attention of readers. You can either compile questions that you've received since the last edition or you can summarize some generic questions that are frequently asked about your organization.

A listing of names and titles of managers in your organization is a good way to give your newsletter a personal touch. If your organization is small, you may want to list the names of all employees.

If you have any prices of standard products or services, you can include a listing of those here. You may want to refer your readers to any other forms of communication that you've created for your organization.

You can also use this space to remind readers to mark their calendars for a regular event, such as a breakfast meeting for vendors every third Tuesday of the month, or a biannual charity auction.

If space is available, this is a good place to insert a clip art image or some other graphic.

### Advertising 10

3.5

10.3

36.05cm<sup>2</sup>

### Advertising 11

40.5cm<sup>2</sup>

9cm

4.5

### Advertising 12

8cm

82.4cm<sup>2</sup>

10.3

PLEASE  
PLACE  
STAMP  
HERE

## Marking Key: Validation, Percentages

| Question                                                                                                                                                                                                                                                             | Marking Key                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. <u>Estimate</u> the percentage of advertising on each page and use this to find the mean. Complete the table.                                                                                                                                                     | / 4 marks<br>- Must have estimate for each page (4) in percentage form.                                                                                                                                                                     |
| 2. Calculate the area ( $\text{cm}^2$ ) of each of the advertisements on each page. Show the calculation used and the area of each advertisement in the spread.<br><b>Record the areas in the table and calculate the total area of advertisement for each page.</b> | / 28 marks<br>- Each page allocated 7 marks<br>- For each page;<br>- Must show correct calculations and correct answer for each advertisement (6)<br>- Must show correct calculations to find the total area of advertisement for each page |
| 3. Determine the Area of the whole page showing your calculation.                                                                                                                                                                                                    | / 2 marks<br>- Must show correct calculations to find correct area of the whole page                                                                                                                                                        |
| 4. Determine the percentage of advertising on each page showing your calculations. (Round to 2 decimal places)<br>Then determine the mean percentage advertising for the whole newsletter.                                                                           | / 10 marks<br>- 2 marks per page; must show correct calculation and correct answer<br>- 2 marks for calculation of mean; correct calculations and answer                                                                                    |
| 5. <u>Compare</u> your estimate to the actual percentage advertising space. How could you have improved your estimate?                                                                                                                                               | / 2 marks<br>- 1 mark for comparing estimate to actual percentage<br>- 1 mark for improvement strategies                                                                                                                                    |
| 6. What fraction of the newsletter remains for us to send out all of the maths news each term?                                                                                                                                                                       | / 2 marks<br>- Must show correct calculations and correct answer to find what fraction remains.                                                                                                                                             |